

Deren Lei

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EDUCATION

University of Southern California (USC)
Computer Science M.S.

LOS ANGELES, CA
Aug 2019 - June 2021

University of California, Santa Barbara (UCSB)
Computer Science B.S.
College of Engineering Honors Program

SANTA BARBARA, CA
Sep 2015 - Jun 2019
GPA: 3.90

RESEARCH EXPERIENCE

UCSB Natural Language Processing Lab
Researcher (Advisor: [William Wang](#))

SANTA BARBARA, CA
Fall 2017 – Present

Publications

- **Deep Reinforcement Learning with Distributional Semantic Rewards for Abstractive Summarization**
Deren Lei*, Siyao Li*, Pengda Qin, William Yang Wang (* equal contribution)
EMNLP-IJCNLP 2019, short paper [available soon]
Description: Deep reinforcement learning in NLP have problems of high variance predictions and sparse reward signals. Using distributed semantic reward to generate more readable and abstractive text.
- **Imposing Label-Relational Inductive Bias for Extremely Fine-Grained Entity Typing**
Wenhan Xiong, Jiawei Wu, **Deren Lei**, Mo Yu, Shiyu Chang, Xiaoxiao Guo and William Yang Wang
NAACL-HLT 2019, long paper [[arXiv:1903.02591](#)]
Description: Recent entity typing tasks moves to more diverse and fine-grained datasets and current learning approaches have contradiction predictions. Improved performances by avoiding inconsistency outputs.
- **Implicit Regularization of Stochastic Gradient Descent in NLP: Observations and Implications**
Deren Lei, Zichen Sun, Yijun Xiao, William Yang Wang
Technical Report, long paper [[arXiv:1811.00659](#)]
Description: Studied the implicit regularization of stochastic gradient descent in practical deep learning systems and its correlation with explicit techniques under noisy and limited samples.

Datasets

- **Multimodal Headline Generation**
Description: Helped AI powered news content platform companies auto-generate attractive smartphone notifications based on contents: headline, story highlights, article, images, categories, etc. Did large dataset collection.
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WORK EXPERIENCE

Amazon

BEIJING, CHINA

Software Engineering Intern (Mentor: Gang Wang)

Summer 2017

- Created an internal tool to help developers to do automatic product page status checking. Saved them more than 5 hours of manual checking the special offer pages per month.
 - Performed more than 100000 data processing to test a softline product recommendation deep learning algorithm that could be applied to Amazon product recommendation system in recent years.
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ACADEMIC PROJECTS

Video Activity Recognition (Advisor: [B.S. Manjunath](#))

Winter 2018

- Helped UCSB Vision Research Lab do behavior prediction by with an emphasis on hospital scenario. Learning via single image frames on AVA dataset using ResNet101.

NoteHub (Developer, Product Manager)

Winter 2018

- An software system with android-end and web-end (mainly focused) for students and professors to share notes in both hand-written and electrical format using Java and Javascript.
 - Contacted Scanbot Team to freely use their API for education purpose for a quarter when it costs €100,000.
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SKILLS

Programming Languages: Python, Java, C/C++, Scala, MATLAB, R, MIPS Assembly, Javascript

Technical Tools: Tensorflow, PyTorch, scikit-learn, Git, emacs, AWS, LaTeX, SQL, Firebase